

# John Evans Dakurah

<https://johnnevansdakurah.github.io> | johnnevansdakurah@gmail.com | +90 50 1080 65 34

## EDUCATION

**Ankara Yıldırım Beyazıt University** | *MSc in Mechanical Engineering (expected)*

- CGPA: 4.00 / 4.00

**Ankara, Turkey** | **Sept, 2025 – June, 2027**

**Gazi University** | *BSc in Automotive Engineering*

- CGPA: 3.56 / 4.00

**Ankara, Turkey** | **Sept, 2021 – June, 2025**

## HONORS & AWARDS

- **Best Graduating Student, Department of Automotive Engineering** — Gazi University, 2025

## TECHNICAL SKILLS

- **Simulation & Modeling:** ANSYS Fluent, ANSYS Workbench (SpaceClaim, DesignModeler), MATLAB, Simulink
- **Programming & Data Analysis:** Python, FORTRAN, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, OriginPro
- **Research Methods:** Computational fluid dynamics (CFD), physics-based numerical modeling, mesh-independence studies, parametric analysis, experimental validation, numerical post-processing, data visualization
- **CAD & Engineering Design:** SolidWorks, CATIA
- **Additional Computing:** JavaScript, HTML, CSS, React, Redux, Bootstrap, jQuery, SASS; linear and logistic regression, decision trees, support vector machines, random forests
- **UAV & Flight-Control Tools:** PyMavlink, MAVProxy, Mission Planner

## RESEARCH EXPERIENCE

**Graduate Research Assistant** | *Prof. Kamil Aslan*

**Ankara, Turkey** | **Sept, 2025 – Present**

- Conduct computational and experimental research in thermal and fluid sciences to investigate fluid flow and heat transfer in energy systems.
- Develop and validate numerical models using mesh-independence studies, parametric analyses, experimental data, and numerical post-processing to evaluate system performance under different operating conditions.
- Contribute to research on thermal-fluid systems and energy-conversion technologies, with applications in fuel cells, internal combustion engines, heat exchangers, and thermal management.

**Undergraduate Research Assistant** | *Prof. Hamit Solmaz*

**Ankara, Turkey** | **Sept, 2022 – Aug, 2024**

- Developed physics-based models and simulations of lithium-ion battery electric, fuel-cell electric, and hybrid electric vehicles using MATLAB and Simulink.
- Developed energy-management systems for hybrid electric vehicles using fuzzy-logic and conventional on-off control, emphasizing energy efficiency and optimization.
- Developed electrochemical models of proton-exchange-membrane (PEM) fuel cells for electric-vehicle applications.

## JOURNAL PUBLICATIONS

- Dakurah, J. E., Solmaz, H., & Kocakulak, T. (2024). Modeling of a PEM Fuel Cell Electric Bus with MATLAB/Simulink. *Automotive Experiences*, 7(2), 252-269. <https://doi.org/10.31603/ae.11471>
- Dakurah, J. E. (Under review). Fuel Cell Electric Vehicles: A Review of Technology, Energy Management Strategies, Challenges, and Future Research Directions. *International Journal of Automotive Engineering and Technologies*.

## TEACHING EXPERIENCE

**Science and Mathematics Tutor** | *Portakal Eğitim Kurumu*

**Ankara, Turkey** | **Feb, 2025 – Present**

- Teach mathematics and science subjects, including physics, chemistry, and biology, to high school students through clear, structured instruction.
- Prepare students for high school examinations using targeted review, practice problems, and progress-based academic support.
- Provide individual guidance and mentoring to help students strengthen subject understanding, problem-solving skills, and study habits.

## WORK EXPERIENCE

**Automotive Engineering Intern** | *Barok Defense*

**İzmir, Turkey** | **July, 2024 – Jan, 2025**

- Performed computational thermal and fluid-flow analyses using ANSYS to evaluate heat-transfer and flow behavior in camera, lighting, communication, and fuel-heating systems for military vehicles.
- Compared alternative system designs, identified promising configurations, and supported design refinement based on simulation results and engineering requirements.
- Built and tested prototypes through experimental thermal, fluid, mechanical, and functional validation to identify performance issues and verify system reliability under demanding operating conditions.

## ACADEMIC PROJECTS

---

- Comparative CFD Analysis of Exhaust Manifold Designs** | *ANSYS Fluent* **Feb, 2026 – May, 2026**
- Compared collector-type and log-type exhaust manifold configurations using computational fluid dynamics to assess flow distribution and thermal behavior.
  - Evaluated velocity, pressure, turbulence, and temperature characteristics to identify design effects on exhaust-flow performance.
- Four-Cylinder Diesel Engine Design & Analysis** | *SolidWorks, ANSYS Workbench* **May, 2024 – June, 2024**
- Designed a four-cylinder, water-cooled diesel engine using SolidWorks.
  - Performed steady-state thermal analysis of the engine using ANSYS Workbench.
- Flight Control Programming of a UAV** | *Python, PyMAVLink, Mission Planner* **Jan, 2024 – March, 2024**
- Implemented MAVLink communication between a UAV and a ground station using Python and PyMAVLink.
  - Implemented MAVLink protocols to send and receive flight information from the UAV.
  - Implemented MAVLink protocols to control UAV movement, including speed, yaw, pitch, roll, and landing.
- Design & Analysis of an M3 Category Articulated Bus** | *SolidWorks, CATIA* **Sept, 2023 – Dec, 2023**
- Developed a 3D model of a 17.1 meter long articulated bus using CATIA
  - Performed structural analyses of the model using SolidWorks.